XRPL Validator — Daily Engineering Report · 2026-05-01

Period: 2026-05-01 00:00 to 24:00 US Eastern (2026-05-01 04:00 to 2026-05-02 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 5 commits (0 with a recorded merge into main; xrpl-validator merge records begin 2026-07-04).

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-05-01): 0.99626 (80 missed of 21,364). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

Merge records begin 2026-07-04; this day's commits are listed by commit date below.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(ffi): silent-divergence detection for tec*/tesSUCCESS mismatches** (`f142a1e` , xrpl-validator) The existing divergences.jsonl only records ter/*tef*/tem/*tel* outcomes — our shim returning tesSUCCESS or tec* is treated as agreeing with the network even when the network actually had a different result. That blind spot mattered after the ltCHILD fix: the residual sequence-tracking divergences (29 in 24h, all OfferCreate/TicketCreate) cluster in single- ledger bursts (e.g. L103934068, 27 from one sender …
- **feat(ffi): expose silent divergences in /api/engine + watch_engine.py** (`0431e77` , xrpl-validator) Counters parallel to live_diverged_by_type: - live_apply_silent_diverged: total - silent_diverged_by_pair: "{tx_type}/{our_ter}->{net_ter}" → count - silent_diverged_samples: rolling last-50 buffer for the dashboard
- **fix(ffi): set header.closeTime so view.parentCloseTime() isn't always epoch** (`9af0b0a` , xrpl-validator) OpenView's open_ledger constructor (OpenView.cpp:105) sets info_.parentCloseTime = base_->info().closeTime — it READS base.closeTime, not base.parentCloseTime. We were setting only header.parentCloseTime; header.closeTime defaulted to 0. Result: view.parentCloseTime() returned the epoch for every tx, and every libxrpl check that consults it broke in a category-specific way: - OfferCreate's hasExpired(view, …
- **feat(ffi): mutation-list divergence detection (BF6C928F class)** (`f0de666` , xrpl-validator) Third detection lane after regular and silent: same TER on both sides, different sets of state mutations. Catches cases where our shim and network agreed the tx applied successfully (or hit the same tec*) but the AffectedNodes the tx ended up touching differ.

Docs and operations

- **chore(watch_engine): remove Live TX Feed block** (`3f7768d`, xrpl-validator)

Independence program

No independence record for UTC day 2026-05-01; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
05-01 13:33	—	xrpl-validator	`f142a1e`	feat(ffi): silent-divergence detection for tec*/tesSUCCESS mismatches
05-01 13:47	—	xrpl-validator	`0431e77`	feat(ffi): expose silent divergences in /api/engine + watch_engine.py
05-02 01:58	—	xrpl-validator	`9af0b0a`	fix(ffi): set header.closeTime so view.parentCloseTime() isn't always epoch
05-02 02:11	—	xrpl-validator	`f0de666`	feat(ffi): mutation-list divergence detection (BF6C928F class)
05-02 02:12	—	xrpl-validator	`3f7768d`	chore(watch_engine): remove Live TX Feed block

Times in UTC. A commit appears on the day it was committed to main or merged into it.